

**SIMATSENGINEERING
ASSESSMENT TOOL 3**

COURSENAME:ArtificialIntelligence

COURSECODE:CSA17

COURSEOUTCOMECOVERED

CO3:Analyzeknowledgerepresentationandreasoningusingpropositionallogic,FOL,inference,resolution, chaining methods, knowledge representation to perform logical reasoning for drawing validconclusions and decision making. (BL4)

AssessmentToolWeightage:20%

QUESTIONS:5

TimeDuration:1Hour
TotalMarks:30

AnnexureA

ResolutionAlgorithmImplementation

CodeofConduct:

I S.SANJAI (192421121)certifythatthissubmissionismyoriginalworkandthatIhaveadheredtothe guidelines specified for this assessment.

Iunderstandthatanyviolationofacademicintegrityruleswillresultindisciplinaryaction.

Signatureofthe student:

Sanjai S

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm. Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

1 –RainandWetGround

PropositionalLogic

Let:

- **R**=Itisraining
- **W**=Groundis wet

Knowledge Base:

$R \rightarrow W$
R

Goal:

W

CNF Conversion

$R \rightarrow W$ becomes:

$\neg R \vee W$

NegatedGoal:

$\neg W$

Python Program:

```
print("ResolutionAlgorithm-RainandWetGround") # CNF

Clauses
C1="¬RVW" C2 =
"R"
C3="¬W"

print("C1=",C1)
print("C2 =", C2)
print("NegatedGoal=",C3)

# Resolution
print("\nResolve¬RVWandR")
print("Result: W")

print("\nResolve W and ¬W")
print("Result:□(EmptyClause)")

print("\nConclusion:Groundiswet-PROVED")
```

Output:

```
Resolution Algorithm
Question 1: Rain and Wet Ground

Knowledge Base:
C1 =  $\neg R \vee W$ 
C2 = R

Negated Goal:
C3 =  $\neg W$ 

Step 1: Resolve C1 and C2
C1 =  $\neg R \vee W$ 
C2 = R
Resolving  $\neg R$  and R
Result: W

Step 2: Resolve W and C3
W and  $\neg W$ 
Result:  $\square$  (Empty Clause)

Conclusion:
Empty Clause  $\square$  is obtained.
Therefore, the goal 'The ground is wet' is PROVED.
```

Final Conclusion:

Since the **Empty Clause** ($\square$) is obtained, the negation of the goal produces a contradiction. Therefore, it is proved that the ground is wet.

2 – Student Assignment Submission

Propositional Logic

Let:

- **S**=Rahul submitted the assignment
- **M**=Rahul receives internal marks

Knowledge Base:

$S \rightarrow M$
S

Goal:

M

CNF Conversion

$S \rightarrow M$ becomes:

$\neg S \vee M$

Negated Goal:

$\neg M$

Python Program:

```
print("ResolutionAlgorithm-StudentAssignmentSubmission") #

CNF Clauses
C1="¬SVM" C2 =
"S"
C3="¬M"

print("C1=",C1)
print("C2 =", C2)
print("NegatedGoal=",C3)

# Resolution
print("\nResolve¬SVMandS")
print("Result: M")

print("\nResolve M and ¬M")
print("Result:□(EmptyClause)")

print("\nConclusion:Rahulreceivesinternalmarks-PROVED")
```

Output:

```
Resolution Algorithm - Student Assignment Submission
C1 = ¬S ∨ M
C2 = S
Negated Goal = ¬M

Resolve ¬S ∨ M and S
Result: M

Resolve M and ¬M
Result: □ (Empty Clause)

Conclusion: Rahul receives internal marks - PROVED
|
```

Conclusion:

Since the **Empty Clause(□)** is obtained, **Rahul receives internal marks is proved.**

3 –Library Membership

Propositional Logic

Let:

- **L**=Priya is a library member
- **B**=Priya can borrow books

Knowledge Base:

$L \rightarrow B \vee L$

Goal:

B

CNF Conversion

$L \rightarrow B$ becomes:

$\neg L \vee B$

NegatedGoal:

$\neg B$

Python Program:

```
print("ResolutionAlgorithm-LibraryMembership")

#CNFClauses
C1="¬LVB"  C2
= "L"
C3="¬B"

print("C1=",C1)
print("C2 =", C2)
print("NegatedGoal=",C3)

# Resolution
print("\nResolve¬LVBandL")
print("Result: B")

print("\nResolve B and ¬B")
print("Result:□(EmptyClause)")

print("\nConclusion:Priyacanborrowbooks-PROVED")
```

Output:

```
Resolution Algorithm - Library Membership
C1 = ¬L ∨ B
C2 = L
Negated Goal = ¬B

Resolve ¬L ∨ B and L
Result: B

Resolve B and ¬B
Result: □ (Empty Clause)

Conclusion: Priya can borrow books - PROVED
```

Conclusion

Since the **Empty Clause**($\square$) is obtained, **Priya can borrow books is proved.**